MODULE 44 · DOMAIN 5 - OPERATIONAL EXCELLENCE AND AUTOMATION

Deployment Strategies and Rollback

Expose a release to as few users as possible until it proves itself, and make going back faster than going forward.

DOMAIN 5 of 5 EXAM WEIGHT 20% MODULE 44 of 52

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Every strategy trades **blast radius** and **rollback speed** against cost and time to release.

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The strategies

STRATEGY	BLAST RADIUS	ROLLBACK	EXTRA COST
All at once	Everyone	Redeploy old	None
Rolling	Grows by batch	Roll back batches	None or one batch
Blue/green	Everyone at switch	Switch back, instant	Double capacity
Canary	Small slice first	Shift slice back	Small
Linear	Equal steps	Shift back	Small

ON THE EXAM

Smallest exposure: canary. Fastest return: blue/green. Cheapest: rolling.

Strategy by compute

COMPUTE	FEATURE	STRATEGIES
Lambda	Alias weights with CodeDeploy	Canary, linear, all at once
ECS	Built-in deployment controller	Rolling, blue/green, canary, linear
EC2 Auto Scaling	Instance refresh	Rolling with checkpoints
EC2	CodeDeploy	In-place or blue/green
API Gateway	Canary release on a stage	Canary

ON THE EXAM

ECS gained built-in blue/green in July 2025 and canary and linear in October 2025.

ECS: built-in or CodeDeploy?

BUILT-IN ECS DEPLOYMENTS

Default for new services

- Blue/green, canary, linear
- Service Connect, multiple target groups
- Bake time before old is retired
- Alarm and hook rollback

CODEDEPLOY FOR ECS

Still supported

- Existing pipelines built on it
- Releases coordinated with CodeDeploy
- Established governance
- No Service Connect

THE TIEBREAKER

New ECS service, or Service Connect: built-in. Existing CodeDeploy process: keep it.

Automatic rollback triggers

WHERE	TRIGGER
CodeDeploy	CloudWatch alarms, failed hooks
ECS built-in	Alarms, lifecycle hooks, bake time
ECS rolling	Deployment circuit breaker
Instance refresh	Alarms, failed health checks
CloudFormation	Rollback triggers with monitoring time

ON THE EXAM

A strategy without an automatic trigger still depends on someone noticing.

Database changes: expand and contract

01
Expand
Add new columns or tables only

02
Deploy app
Works with old and new schema

03
Migrate
Backfill and switch reads

04
Contract
Remove old columns later

ON THE EXAM

Rollback only works if the old version can still use the database.

Reading the scenario

WHEN THE QUESTION SAYS	IT IS TESTING
Small percentage first, revert on errors	Canary with alarm rollback
Switch back instantly, test full stack first	Blue/green
Tasks keep failing, deployment hangs	ECS deployment circuit breaker
New AMI across an Auto Scaling group	Instance refresh, checkpoints
Rollback broke on a schema change	Expand and contract
ECS with Service Connect	Built-in ECS deployments

ON THE EXAM

Match the risk to the strategy, then the compute to the feature.

> Larkspur Payments runs card authorisation in a Lambda function. A new version should take 10% of traffic for 10 minutes, then the rest.
>
> If the error alarm fires during that time, traffic must return to the old version automatically.

What should the solutions architect implement?

A. Publish the new version and point callers at $LATEST

B. CodeDeploy with a Canary10Percent10Minutes configuration on the function alias, with the error alarm attached

C. Route 53 weighted records for two functions

D. A rolling deployment of the function

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A $LATEST
All traffic at once, no rollback.

B CodeDeploy canary on alias
10% for 10 minutes, alarm rolls back.

C Route 53 weights
DNS caching, and no automatic rollback.

D Rolling
Lambda has no instances to roll through.

THE SIGNAL

Lambda, small slice first, automatic revert: CodeDeploy canary on the alias.

> Larkspur Payments is launching a new ECS service on Fargate that talks to other services through ECS Service Connect.
>
> It wants canary releases, a bake time before the old revision is removed, and rollback on alarms, with as few extra services as possible.

What should the solutions architect use?

A. CodeDeploy blue/green for ECS

B. Rolling updates with minimum healthy percent of 50

C. The ECS built-in deployment controller with a canary strategy, bake time and CloudWatch alarms

D. Two ECS services behind Route 53 weighted records

◆ PAUSE AND DECIDE

What should the solutions architect use?

A CodeDeploy

Does not support Service Connect, and adds a service.

B Rolling

No traffic slice, no bake time.

C ECS built-in canary

Canary, bake time, alarm rollback, Service Connect.

D Route 53 split

Custom, imprecise, no automatic rollback.

THE SIGNAL

New ECS service with Service Connect: the built-in deployment controller.

> Larkspur Payments uses ECS rolling updates for an internal service. A release with a bad image caused new tasks to fail health checks again and again.
>
> The deployment kept retrying for three hours before anyone noticed and rolled back by hand.

What should the solutions architect enable?

A The ECS deployment circuit breaker with rollback enabled

B A longer health check grace period

C Blue/green deployments with twice the capacity

D A larger minimum healthy percent

◆ PAUSE AND DECIDE

What should the solutions architect enable?

A **Circuit breaker**
Stops a failing rollout and returns to the last good revision.

B **Longer grace**
Delays detection further.

C **Blue/green**
A bigger change than the problem needs.

D **Minimum healthy**
Protects capacity, does not stop retries.

THE SIGNAL

ECS tasks keep failing during a rolling update: deployment circuit breaker.

> Larkspur Payments runs 200 EC2 instances in an Auto Scaling group. A new AMI must replace them gradually, pausing at 20% for validation.
>
> If the latency alarm fires during the rollout, the group must return to the previous launch template automatically.

What should the solutions architect use?

A Terminate instances in batches by hand

B Create a second Auto Scaling group and switch DNS

C CodeDeploy in-place deployment

D Instance refresh with a checkpoint at 20%, alarms, and auto rollback enabled

◆ PAUSE AND DECIDE

What should the solutions architect use?

A **Manual batches**
No checkpoint logic, no rollback.

B **Second group**
Blue/green; not gradual, and more to build.

C **CodeDeploy in place**
Deploys application code, not a new AMI.

D **Instance refresh**
Gradual, pauses at checkpoints, rolls back on alarm.

THE SIGNAL

New AMI across an Auto Scaling group, gradual, with rollback: instance refresh.

> Larkspur Payments used blue/green for a release that renamed a database column. When an error forced a switch back, the old version failed because the column it expected no longer existed.
>
> Future releases must keep the switch back working.

What should the solutions architect change?

A Restore a database snapshot during each rollback

B Use expand and contract: add new columns first, keep old ones until no rollback is needed

C Deploy all at once at night

D Give blue and green separate databases

◆ PAUSE AND DECIDE

What should the solutions architect change?

A Snapshot restore

Loses transactions written since the snapshot.

B Expand and contract

Both versions work with the schema during the release.

C All at once

Makes rollback harder, not easier.

D Separate databases

Data diverges; payments split across two stores.

THE SIGNAL

Rollback broken by a schema change: expand and contract.

> Larkspur Payments' settlement service runs on EC2 behind an Application Load Balancer and is released twice a year. Each release must be tested end to end in production conditions before any customer traffic reaches it.
>
> If a problem appears, the return must be immediate. Doubling capacity for a day is acceptable.

Which strategy should the solutions architect choose?

A Canary, 5% of traffic for an hour

B Rolling, one instance at a time

C Blue/green with CodeDeploy, testing the green environment before shifting traffic

D All at once during a maintenance window

◆ PAUSE AND DECIDE

Which strategy should the solutions architect choose?

A **Canary**
Customers reach the new version before full testing.

B **Rolling**
Mixed versions, slow rollback.

C **Blue/green**
Test green fully, switch, switch back instantly.

D **All at once**
No pre-traffic test, no fast return.

THE SIGNAL

Test before any customer traffic, instant return, double capacity allowed: blue/green.

What to carry out of this module

- **Canary** for smallest exposure; **blue/green** for fastest return.
- **Lambda**: CodeDeploy on aliases. **ASG**: instance refresh.
- **ECS built-in** blue/green, canary and linear for new services.
- **Automatic triggers**: alarms, hooks, circuit breaker.
- **Expand and contract** keeps rollback possible.

> ON THE EXAM
>
> Match the risk to the strategy, then the compute to the feature.